

MTH-ALG, Algebra Problem Sets Solutions

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§1 Problem Set 1

Exercise 1.1. Let

$$G = \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/20\mathbb{Z}. \quad (1.1)$$

- (a) How many elements of G have order 60?
- (b) How many cyclic subgroups of G have order 30?
- (c) How many injective homomorphisms

$$f : \mathbb{Z}/12\mathbb{Z} \longrightarrow G \quad (1.2)$$

are there?

Solution. (a) We shall prove something first:

Claim 1.1: If $a \in G$ has order n , and $b \in H$ has order m , then $\text{ord}((a, b)) = \text{lcm}(n, m)$ in $G \times H$.

Proof. Suppose $(a, b)^k = (e, e)$. Then $a^k = b^k = e$. This gives us that $n \mid k$ and $m \mid k$. Hence,

$$\text{lcm}(n, m) \mid k. \quad (1.3)$$

Furthermore,

$$(a, b)^{\text{lcm}(n, m)} = (a^{\text{lcm}(n, m)}, b^{\text{lcm}(n, m)}) = (e, e). \quad (1.4)$$

Hence, $\text{ord}((a, b)) = \text{lcm}(n, m)$. $\square$

Now, (a, b) will have order 60 if $\text{lcm}(\text{ord}(a), \text{ord}(b)) = 60$. Since $\text{ord}(a) \in \{1, 2, 3, 6\}$ and $\text{ord}(b) \in \{1, 2, 4, 5, 10, 20\}$, $\text{lcm}(\text{ord}(a), \text{ord}(b)) = 60$ can only happen in the following cases:

- $\text{ord}(a) = 3, \text{ord}(b) = 20$. Then the choices for b are all the numbers coprime to 20, so we have $\varphi(20) = 8$ choices. The choices for a are 2, 4. Hence there are $8 \times 2 = 16$ possibilities.
- $\text{ord}(a) = 3, \text{ord}(b) = 20$. Then the choices for b are all the numbers coprime to 20, so we have $\varphi(20) = 8$ choices. The choices for a are 1, 5. Hence there are $8 \times 2 = 16$ possibilities.

Hence, there are $16 + 16 = 32$ elements of order 60.

- (b) Let's first calculate how many elements (a, b) are there of order 30. As before, the possibilities are:

- $\text{ord}(a) = 3, \text{ord}(b) = 10$. Then the choices for b are multiples of 2 by all the numbers coprime to 10, so we have $\varphi(10) = 4$ choices. The choices for a are 2, 4. Hence there are $4 \times 2 = 8$ possibilities.

- $\text{ord}(a) = 6, \text{ord}(b) = 10$. Then the choices for b are multiples of 2 by all the numbers coprime to 10, so we have $\varphi(10) = 4$ choices. There are $\varphi(6) = 2$ choices for a . Hence there are $4 \times 2 = 8$ possibilities.
- $\text{ord}(a) = 6, \text{ord}(b) = 5$. Then the choices for b are multiples of 4 by all the numbers coprime to 5, so we have $\varphi(5) = 4$ choices. There are $\varphi(6) = 2$ choices for a . Hence there are $4 \times 2 = 8$ possibilities.

Hence, there are $8 + 8 + 8 = 24$ numbers of order 30. A subgroup of order 30 consists of $\varphi(30) = 8$ generators. Hence, the number of subgroups of order 30 is $24/8 = 3$.

(c) Note that f is completely fixed by $f(\bar{1})$. For f to be injective, $f(\bar{1})$ has to be an element of order 12. Let's now calculate how many elements (a, b) are there of order 12. The possibilities are:

- $\text{ord}(a) = 3, \text{ord}(b) = 4$. Then the choices for b are multiples of 5 by all the numbers coprime to 4, so we have $\varphi(4) = 2$ choices. The choices for a are 2, 4. Hence there are $2 \times 2 = 4$ possibilities.
- $\text{ord}(a) = 6, \text{ord}(b) = 4$. Then the choices for b are multiples of 5 by all the numbers coprime to 4, so we have $\varphi(4) = 2$ choices. There are $\varphi(6) = 2$ choices for a . Hence there are $2 \times 2 = 4$ possibilities.

Hence, there are $4 + 4 = 8$ elements of order 12. As a result, there are 12 injective homomorphisms $f : \mathbb{Z}_{12} \rightarrow G$.

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Exercise 1.2. Let $(G_i)_{i=1}^n$ be a family of normal subgroups of G such that

$$G = \left\langle \bigcup_{i=1}^n G_i \right\rangle. \quad (1.5)$$

Show that G is isomorphic to the direct product of the G_i if and only if one of the following conditions holds:

1. For every i ,

$$G_i \cap \left\langle \bigcup_{j \neq i} G_j \right\rangle = \{e\}. \quad (1.6)$$

2. Every element $g \in G$ can be written uniquely as a product

$$g = g_1 g_2 \cdots g_n, \quad g_i \in G_i. \quad (1.7)$$

Solution. We will show the equivalence of the following 3 statements:

1. The map $f : G_1 \times \cdots \times G_n \rightarrow G$ defined by

$$f(g_1, g_2, \dots, g_n) = g_1 g_2 \cdots g_n \quad (1.8)$$

is an isomorphism.

2. Every element $g \in G$ can be written uniquely as a product

$$g = g_1 g_2 \cdots g_n, \quad g_i \in G_i. \quad (1.9)$$

3. For every i ,

$$G_i \cap \left\langle \bigcup_{j \neq i} G_j \right\rangle = \{e\}. \quad (1.10)$$

Claim 1.2: Assuming 2, if $g_i \in G_i$ and $g_j \in G_j$ for $i \neq j$, then $g_i g_j = g_j g_i$.

Proof. WLOG, assume $i < j$. Since G_i is normal, $g_j g_i g_j^{-1} \in G_i$, so let's call this element g'_i . Therefore,

$$g_j g_i = g'_i g_j. \quad (1.11)$$

Similarly, since G_j is normal, $g_i^{-1} g_j g_i \in G_j$, so let's call this element g'_j . Therefore,

$$g_j g_i = g_i g'_j. \quad (1.12)$$

But this gives two different representations of the element $g_j g_i$:

$$g_j g_i = e \cdots e \underbrace{g'_i}_{i\text{-th slot}} e \cdots e \underbrace{g_j}_{j\text{-th slot}} e \cdots e = e \cdots e \underbrace{g_i}_{i\text{-th slot}} e \cdots e \underbrace{g'_j}_{j\text{-th slot}} e \cdots e. \quad (1.13)$$

By the uniqueness, we have $g_i = g'_i$ and $g_j = g'_j$, proving that $g_j g_i = g_i g_j$. $\square$

Claim 1.3: Assuming 3, if $g_i \in G_i$ and $g_j \in G_j$ for $i \neq j$, then $g_i g_j = g_j g_i$.

Proof. Consider the element $x := g_i g_j g_i^{-1} g_j^{-1}$. By the normality of G_j , $g_i g_j g_i^{-1} \in G_j$, so that $x \in G_j$. Similarly, by the normality of G_i , $g_j g_i^{-1} g_j^{-1} \in G_i$, so that $x \in G_i$. Since the intersection $G_i \cap G_j = \{e\}$, we can conclude that $x = e$, i.e. $g_i g_j = g_j g_i$. $\square$

(1 $\Rightarrow$ 2) Trivial.

(2 $\Rightarrow$ 3) Let $x \in G_i \cap \left\langle \bigcup_{j \neq i} G_j \right\rangle$. Since $x \in G_i$, it can be uniquely represented as

$$x = e \cdots e \underbrace{x}_{i\text{-th slot}} e \cdots e. \quad (1.14)$$

Since $x \in \langle \bigcup_{j \neq i} G_j \rangle$, x can be written as a string of elements of G_j . Since elements from different G_j 's commute, we can group the elements from the same G_j 's and then form

$$x = g_1 g_2 \cdots g_{i-1} e g_{i+1} \cdots g_n. \quad (1.15)$$

by uniqueness, we must have that all g_j 's must be e , and also $x = e$. Hence, $G_i \cap \langle \bigcup_{j \neq i} G_j \rangle = \{e\}$.

(3 $\Rightarrow$ 1) First, we'll show that f is a homomorphism.

$$f(g_1, \dots, g_n) f(g'_1, \dots, g'_n) = (g_1 \cdots g_n) (g'_1 \cdots g'_n). \quad (1.16)$$

Since elements from different G_i 's commute, this can be rewritten as

$$(g_1 \cdots g_n) (g'_1 \cdots g'_n) = (g_1 g'_1) (g_2 g'_2) \cdots (g_n g'_n) = f(g_1 g'_1, \dots, g_n g'_n), \quad (1.17)$$

i.e. f is a homomorphism. f is surjective indeed: since $G = \langle \bigcup_{i=1}^n G_i \rangle$, every element $g \in G$ can be written as a string of elements of G_i 's. Since elements from different G_i 's commute, we can group the elements from the same G_i 's and then form

$$g = g_1 g_2 \cdots g_n, \quad \text{where } g_i \in G_i \text{ for each } i. \quad (1.18)$$

Now, suppose $(g_1, \dots, g_n) \in \text{Ker } f$. Then

$$e = g_1 g_2 \cdots g_n. \quad (1.19)$$

From this, we can write

$$\begin{aligned} g_i^{-1} &= g_i^{-1} g_1 g_2 \cdots g_n = g_1 g_2 \cdots g_{i-1} (g_i^{-1} g_i) g_{i+1} \cdots g_n \\ &= g_1 g_2 \cdots g_{i-1} g_{i+1} \cdots g_n \in \left\langle \bigcup_{j \neq i} G_j \right\rangle. \end{aligned} \quad (1.20)$$

(Here the reordering is possible because g_i^{-1} commutes with the other g_j 's.) Since $G_i \cap \langle \bigcup_{j \neq i} G_j \rangle = \{e\}$, this forces that $g_i = e$. This is true for each i . Hence, $\text{Ker } f$ is trivial, and hence it's injective. This completes the proof. $\blacksquare$

Exercise 1.3. Let G be a group, and define

$$\Delta = \{(x, x) \mid x \in G\}. \quad (1.21)$$

- (a) Show that Δ is a subgroup of $G \times G$.
- (b) Prove that Δ is a normal subgroup of $G \times G$ if and only if G is abelian. Moreover, in this case, show that the quotient group $(G \times G)/\Delta$ is isomorphic to G .

Solution. (a) Take $(x, x), (y, y) \in \Delta$. Then

$$(x, x) * (y, y)^{-1} = (x, x) * (y^{-1}, y^{-1}) = (xy^{-1}, xy^{-1}) \in \Delta. \quad (1.22)$$

Therefore, Δ is a subgroup of $G \times G$.

- (b) All subgroups of an abelian group are normal. So if G is abelian, so is $G \times G$, and hence Δ is a normal subgroup.

Suppose Δ is a normal subgroup. Then given any $(g, h) \in G \times G$,

$$(g, h) * (x, x) * (g, h)^{-1} = (g, h) * (x, x) * (g^{-1}, h^{-1}) = (gxg^{-1}, h x h^{-1}). \quad (1.23)$$

This is in Δ , by hypothesis that Δ is normal. Therefore,

$$gxg^{-1} = h x h^{-1} \quad (1.24)$$

Since this holds for every $g, x, h \in G$, we choose $h = e$, so we have

$$gxg^{-1} = x \implies gx = xg, \quad (1.25)$$

proving that G is abelian. Suppose G is abelian, then we define

$$\begin{aligned} f : G \times G &\rightarrow G \\ (x, y) &\mapsto xy^{-1}. \end{aligned} \quad (1.26)$$

This is a homomorphism, because

$$f((g, h) * (x, y)) = f(gx, hy) = (gx)(hy)^{-1} = gxy^{-1}h^{-1} \quad (1.27)$$

$$= (gh^{-1})(xy^{-1}) = f(g, h)f(x, y). \quad (1.28)$$

Therefore, f is a homomorphism. Now,

$$(x, y) \in \text{Ker } f \iff xy^{-1} = e \iff x = y. \quad (1.29)$$

Therefore, $\text{Ker } f = \Delta$. f is surjective, since any $g \in G$ is $f(g, e) = g$. Hence, by the first isomorphism theorem,

$$\text{im } f \cong \frac{\text{dom } f}{\text{Ker } f} \implies G \cong \frac{G \times G}{\Delta}. \quad (1.30)$$

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Exercise 1.4. A subgroup M of a group G is called *maximal* if $M \neq G$ and for every subgroup L of G ,

$$M \subsetneq L \subseteq G \implies L = G. \quad (1.31)$$

- (a) Let K, M be subgroups of a group G such that $K \trianglelefteq G$ and $K \subseteq M$. Prove that M is a maximal subgroup of G if and only if M/K is a maximal subgroup of G/K .

- (b) Prove that in a finite group G , every proper subgroup $H \neq G$ is contained in some maximal subgroup of G .
- (c) Prove that in a finite abelian group different from the trivial group, the maximal subgroups are precisely those of prime index.
- (d) Is the previous point true if we remove the assumption that the group is abelian?

Solution. (a) Let $\pi : G \rightarrow G/K$ denote the projection map. Suppose M is a maximal subgroup of G . Assume for the sake of contradiction that M/K is **NOT** maximal. Then there exists another subgroup

$$\frac{M}{K} \subsetneq X \subsetneq \frac{G}{K}. \quad (1.32)$$

Now consider $\pi^{-1}(X)$.

Claim 1.4: If $f : G_1 \rightarrow G_2$ is a homomorphism, and $H_2 \subseteq G_2$ is a subgroup, then $f^{-1}(H_2) \subseteq G_1$ is a subgroup.

Proof. Take $x, y \in f^{-1}(H_2)$. Then $f(x), f(y) \in H_2$. Since $H_2 \subseteq G_2$ is a subgroup,

$$f(xy^{-1}) = f(x)f(y)^{-1} \in H_2, \quad (1.33)$$

proving that $xy^{-1} \in f^{-1}(H_2)$. So $f^{-1}(H_2) \subseteq G_1$ is a subgroup. $\square$

Now, $\pi^{-1}(X)$ is a subgroup of G , that contains $\pi^{-1}(M/K) = M$. Hence, either $\pi^{-1}(X) = M$ or $\pi^{-1}(X) = G$.

$$\pi^{-1}(X) = \{g \in G \mid gK \in X\}. \quad (1.34)$$

If $\pi^{-1}(X) = G$, then $X = G/K$; and if $\pi^{-1}(X) = M$, then $X = M/K$; both of which contradict (1.32).

Conversely, suppose M/K is a maximal subgroup in G/K . Assume for the sake of contradiction that M is **NOT** maximal. Then there exists another subgroup

$$M \subsetneq H \subsetneq G. \quad (1.35)$$

Since image of a subgroup is also a subgroup, we have $\frac{M}{K} \subset \frac{H}{K} \subset \frac{G}{K}$. By the maximality of M/K , either $H/K = M/K$ or $H/K = G/K$.

Claim 1.5: For any subgroup $L \supseteq K$, $\pi^{-1}(L/K) = L$.

Proof. It's clear that $L \subseteq \pi^{-1}(L/K)$. Now take any $g \in \pi^{-1}(L/K)$. Then $gK \in L/K$.

$$gK = lK \text{ for some } l \in L. \quad (1.36)$$

Therefore, $l^{-1}g \in K \subseteq L$. Hence, $g = l(l^{-1}g) \in L$. As a result, $L \supseteq \pi^{-1}(L/K)$. $\square$

If $H/K = M/K$, by taking π^{-1} , we get $H = K$; and if $H/K = G/K$, then $H = G$; both of which contradict (1.35).

- (b) Let H be a proper subgroup of a finite group G . We construct a sequence of subgroups

$$H =: H_0 \subsetneq H_1 \subsetneq H_2 \subsetneq \cdots \quad (1.37)$$

If H_i is maximal, then we are done, since H is contained in a maximal subgroup! Otherwise, there exists some $g_{i+1} \in G - H_i$ such that

$$H_i \subsetneq H_{i+1} := \langle H_i, g_{i+1} \rangle \subsetneq G. \quad (1.38)$$

If there is no such g_{i+1} , then H_i is indeed maximal, because adding one element to H_i and making a group with it makes it the whole G .

Now, since G is a finite group, this sequence can't go on forever. Therefore, it has to stop somewhere, and thus we find a maximal subgroup that contains H .

- (c) Let M be a subgroup of prime index, i.e. $[G : M] = p$. Let's pick a subgroup L with

$$M \subseteq L \subseteq G. \quad (1.39)$$

Then

$$[G : M] = [G : L][L : M]. \quad (1.40)$$

(This follows from the fact that for finite groups $[G : H] = \frac{|G|}{|H|}$.) Since $[G : M]$ is a prime, it forces either $[G : L] = 1$, or $[L : M] = 1$; consequently $G = L$ or $L = M$. Therefore, M is maximal.

Now suppose M is a maximal subgroup in a finite abelian group G . Since G is abelian, M is normal, so we can form the quotient group G/M . By part (a), since M is maximal, we can conclude that $\{\bar{e}\} = M/M$ is a maximal subgroup of G/M .

Claim 1.6: If $\{e\}$ is a maximal subgroup of $G \neq \{e\}$, then $G \cong \mathbb{Z}_p$ for some prime p .

Proof. Adding one single non-identity element makes it the whole group, so for every $a \in G$ with $a \neq e$,

$$\langle a \rangle = G. \quad (1.41)$$

Hence, G is cyclic. If $|G|$ was infinite, then it would have subgroups other than the trivial one. But as $\{e\}$ is a maximal subgroup, it's the only subgroup. Therefore, $|G|$ can't be infinite.

Now, if $|G|$ was composite, say $|G| = mn$, then $a^{mn} = e$, proving a^m has order n . This gives rise to a subgroup of size n , namely $\langle a^m \rangle$, which is a contradiction! Therefore, $|G|$ is prime. $\square$

Now, $\{\bar{e}\} = M/M$ is a maximal subgroup of G/M , proves that $|G/M| = p$. Therefore,

$$M \subsetneq G \text{ is maximal} \iff [G : M] \text{ is prime.} \quad (1.42)$$

- (d) No. We can consider the group $G = A_4$ of alternating group of degree 4, which contains all even permutations of $\{1, 2, 3, 4\}$. The subgroup

$$H = \langle (1 \ 2 \ 3) \rangle \quad (1.43)$$

has size 3, so that $[G : H] = 2^2$ is not a prime. However, H is maximal. Indeed, if there exists another subgroup

$$H \subsetneq K \subsetneq G, \quad (1.44)$$

then $|K|$ divides $|G| = 12$, and $|H| = 3$ divides $|K|$, forces $|K| = 6$. If K contains a product of two 2-cycles, e.g. $(1 \ 2)(3 \ 4)$, then

$$(1 \ 2 \ 3)(1 \ 2)(3 \ 4) = (1 \ 3 \ 4) \in K \quad (1.45)$$

Since K is a group, it contains the inverse $(1 \ 4 \ 3)$ as well. These are all the 6 elements of K . But then

$$(1 \ 2)(3 \ 4)(1 \ 2 \ 3) = (2 \ 4 \ 3) \quad (1.46)$$

that gives one more element in K , which is a contradiction!

Therefore, K doesn't contain any product of two 2-cycles, i.e. the only non-identity elements in K are 3-cycles. But every 3-cycle comes with its inverse as well, and no 3-cycle is its own inverse. Therefore, $|K|$ can't be even! Thus we arrive at a contradiction. Hence, H is maximal.

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Exercise 1.5. Let p be a prime number.

- (a) Determine all elements of order dividing p in S_n . Denote by a_n the number of such elements and derive a formula for a_n .
- (b) Prove the identity

$$\sum_{n=0}^{\infty} \frac{a_n}{n!} x^n = e^{x + \frac{x^p}{p}}. \quad (1.47)$$

Solution. (a) Every element $\sigma \in S_n$ can be decomposed into disjoint cycles. The disjoint cycles commute. Now, if $\text{ord } \sigma \mid p$, then we must have $\sigma^p = e$. This will happen if and only if $c_i^p = e$ for each disjoint cycle c_i in σ .

Now, for a cycle c_i , $c_i^p = e$ happens if the length of the cycle is either 1 or p . Therefore, a_n will be the number of ways we can decompose $\{1, 2, \dots, n\}$ into cycles of length 1 or p .

Now, suppose $\sigma \in S_n$ consists of m disjoint cycles of length p , and the other $n - mp$ elements are fixed points (i.e. cycles of length 1). Here, m ranges from 0 to $\left\lfloor \frac{n}{p} \right\rfloor$. Therefore,

$$a_n = \sum_{m=0}^{\left\lfloor \frac{n}{p} \right\rfloor} \text{number of } \sigma \text{'s with } m \text{ disjoint cycles of length } p. \quad (1.48)$$

First, we can choose the numbers involved in the p -cycles in $\binom{n}{mp}$ ways. Now, we can partition them into m sets of size p , which can be done in

$$\binom{mp}{p} \cdot \binom{mp-p}{p} \cdot \dots \cdot \binom{p}{p} = \frac{(mp)!}{(p!)^m} \quad (1.49)$$

many ways. Now, each cycles can be permuted in $(p-1)!$ different ways. Finally, the cycles themselves can permute in $m!$ ways among themselves. So we need to divide by that. Therefore, the number of σ 's with m disjoint cycles of length p is

$$\binom{n}{mp} \cdot \frac{(mp)!}{(p!)^m} \cdot ((p-1)!)^m \cdot \frac{1}{m!} = \frac{n!}{p^m m! (n - mp)!}. \quad (1.50)$$

Hence,

$$a_n = \sum_{m=0}^{\left\lfloor \frac{n}{p} \right\rfloor} \frac{n!}{p^m m! (n - mp)!}. \quad (1.51)$$

(b) Using (1.51), we have

$$f(x) := \sum_{n=0}^{\infty} \frac{a_n}{n!} x^n = \sum_{n=0}^{\infty} \sum_{m=0}^{\left\lfloor \frac{n}{p} \right\rfloor} \frac{x^n}{p^m m! (n - mp)!}. \quad (1.52)$$

We can swap the sum into

$$f(x) = \sum_{m=0}^{\infty} \sum_{n=mp}^{\infty} \frac{x^n}{p^m m! (n - mp)!}. \quad (1.53)$$

Suppose $k = n - mp$, so that $n = k + mp$. Then the sum can be rewritten as

$$f(x) = \sum_{m=0}^{\infty} \sum_{k=0}^{\infty} \frac{x^{k+mp}}{p^m m! k!} = \sum_{m=0}^{\infty} \frac{1}{m!} \left(\frac{x^p}{p} \right)^m \sum_{k=0}^{\infty} \frac{x^k}{k!} = e^{\frac{x^p}{p}} e^x = e^{\frac{x^p}{p} + x}. \quad (1.54)$$

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